

$$1) T = 625 \text{ K} \quad t = 500 \text{ dy} \quad L = 600 \mu\text{m}$$

a) oxide thickness?

$$t^*(d) = 6.63 \times 10^{-7} \exp\left(\frac{11949}{T}\right) = 133 \text{ dy} \quad t > t^*$$

$$\delta^*(\mu\text{m}) = 5.1 \exp\left(\frac{-550}{T}\right) = 2.115 \mu\text{m}$$

$$\delta = \delta^* + K_L (t - t^*)$$

$$K_L \left(\frac{\mu\text{m}}{\text{s}}\right) = 7.48 \times 10^4 \exp\left(\frac{-12500}{T}\right) = 0.01542$$

$$\delta = 2.115 + 0.01542 (500 - 133) = \underline{7.773 \mu\text{m}}$$

b) H pickup function

$$C_H (\text{wt. ppm}) = \frac{2f \times \delta \times f_{ox} \times f_{ZrO_2}^0 \times \frac{M_H}{M_O} \times 10^6}{\left(t - \frac{\delta}{PBR}\right) \times \rho_{met-1}}$$

$$f = 0.14 \quad \delta = 7.773 \mu\text{m}$$

$$f_{ZrO_2}^0 = \frac{32}{91.52} = 0.26$$

$$f_{ox} = 5.68 \mu\text{e}$$

$$\frac{M_H}{M_O} = \frac{1}{16}$$

$$PBR = 1.56$$

$$\rho_{met-1} = 6.5 \mu\text{e}$$

$$t = 600 \mu\text{m}$$

$$= \frac{2(0.14)(7.773)(5.68)(0.26)\left(\frac{1}{16}\right) \times 10^6}{\left(600 - \frac{7.773}{1.56}\right)(6.5)}$$

$$= \underline{51.9 \text{ wt. ppm}}$$

2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?

Oxygen diffusion through the oxide layer.

3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?

While hydrogen can diffuse throughout the cladding, H preferentially diffuses down the temperature gradient, and thus collects on the outside of the cladding. The solubility of H is temperature-dependent, being proportional to temperature. Hydrides also preferentially form in areas of tensile stress, which are present on the outside of the cladding. Thus, the higher concentrations of H on the outside of the cladding, combined with lower solubility and tensile stress, lead to preferential hydride formation in this location. Hydrides can embrittle the cladding by impeding dislocation motion, as well as providing stress concentrators.

4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.

Reactivity-initiated accident. Control rod drop/ejection accident. Initial rapid power pulse which leads to a rapid increase in temperature. This temperature increase leads to thermal expansion and a pressure increase inside gas bubbles. This can lead to gap closure, excessive stress on the cladding, departure from nucleate boiling, and potentially fuel fragmentation and dispersal. The type of failure depends on whether the gap is closed, and the amount of energy imparted, along with the pulse width. A RIA occurs very rapidly.

5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?

Loss of coolant accident. Temperature increase from decay heat coupled with a loss of active coolant. Loss of pressure boundary leads to higher pressure inside cladding. Slow temperature rise leads to increased oxidation, potential phase changes of Zr, ballooning, rupture, and/or cladding embrittlement. Once emergency core cooling kicks in, the rapid quench can potentially fracture the embrittled cladding. A LOCA occurs on the timescale of minutes, rather than seconds like a RIA.

6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?

Failure probability increases with burnup, typically due to radiation hardening of the material systems reducing their ductility. The type of failure can change based upon whether the gap is closed or not (PCMI-based or post-DNBR-based), which itself is a function of burnup.

7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?

Improved reaction kinetics with steam, improved cladding properties, improved fuel properties, improved fission product retention.

8. (6 pts) List one near-term ATF concept and why it is of interest.

Cr-coated Zr cladding or Cr₂O₃-doped UO₂. The Cr coating provides a more stable high-temperature oxide phase that can resist high-temperature steam oxidation better than ZrO₂.

9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?

Rapid oxidation, the exothermic reaction from oxidation leads to an increase in temperature, which further accelerates oxidation. If temperatures exceed 863 C, then a phase transition can occur. Oxygen can stabilize the alpha phase to higher temperatures, leaving a brittle shell of oxygen-stabilized alpha phase and a core of plastic beta phase. Above 1200C, all of the cladding is consumed by oxidation.

10. (6 pts) List three examples of limiting phenomena governing LWR operation.

Cladding wear, power to melt, DNBR, etc.

11. (6 pts) What fuel performance and safety impacts does CRUD have?

Limits thermal transport into the coolant as it has a low thermal conductivity and deposits on the cladding. Crud-induced power shifts and modify localized powers, impacting power profiles. Activated corrosion products can deposit on out-of-core surfaces, leading to radiation exposure to personnel during outages.

12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.

Hydrogen water chemistry: lowers the redox potential by limiting the amount of free oxygen in the water. LiOH addition, balances the pH, offsetting the addition of boric acid as a chemical reactivity shim.

13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

Constituent redistribution. Oxygen diffuses down the temperature gradient, leading to near stoichiometric MOX at the pellet rim, and hypostoichiometric MOX inside the fuel. The Pu distribution increases in the columnar grain region up to the central void.

Restructuring. MOX fuel restructures into four regions, central void, columnar grains, large equiaxed grains, and as-fab microstructure. The higher temperatures lead to different mechanisms for microstructural change than LWR fuels.